TRUNKBELAY · RISK REGISTER AND HAZARD ANALYSIS

TrunkBelay risk register and hazard analysis

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	TKB-RSK-001	Version	0.2
Status	Draft	Date	2026-10-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/trunkbelay		
Source	docs/09-risk.md @ b6393c7		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	H6 and H12 controls as decided by Amish (O7, O8), 2026-10-25
0.1	2026-10-25	Amish Chadha	First risk register, preliminary hazard analysis and first-pass design FMEA (kit 1.9.1 upgrade, stage P3)

TrunkBelay risk register and hazard analysis

The top risks are a collar that slides on a wet trunk because the friction is lower than assumed, a person dropped by a wrongly rigged or misplaced collar, and a strength margin at the low end of rescue practice; the top project risk is that no co-design partner runs the friction and drill tests that decide all three. Method: STANDARDS section 23. Likelihood, impact, severity and occurrence are scored 1 (low) to 5 (high); detection 1 (sure to be found before use) to 5 (not found).

> **Safety:** Every hazard below can drop a person from 6 to 25 m. Nobody is lowered with this kit before the collar has been proof-tested at 2.5 kN on cut trunk sections and the drill has been run with a dummy at low height (TKB-BLD-001, section 6). Emergency services are always called as well.

Project risk register

Table 1. Project risks.

#	Risk	Cause	Effect	Likelihood	Impact	Owner (role)	Response
P1	No co-design partner runs the TRL 4 tests	None of the first candidates (training scheme, fire station, climbers' collective) approached yet	The design stays on paper; the deciding unknown (friction) is never measured	3	5	Project lead	Approach the Friends of Coconut Tree training programme and a Kasaragod or Kannur fire station first (TKB-PLN-001)
P2	Wet-bark friction below 0.58 on small trunks	Rubber and nylon on wet, fibrous, possibly algae-covered bark	Option A does not lock on 200 mm trunks; redesign of the lever geometry or pads	2	5	Design lead	Friction test first (BLD first check 1); recorded fallback: longer lever arm, higher-friction pad compound
P3	Concept choice delayed	Options A and B within 10 % (TKB-DEC-001 O5)	Effort split between two anchors	3	2	Project lead	Put O5 to Amish with this upgrade; test both anchors' slip on the same trunk sections if he wishes
P4	Bought rescue gear dearer than the BOM	The BOM's USD 280 descender is a class price; a cited branded device lists at USD 421	PR-009 (USD 850) exceeded	3	3	Project lead	Group purchase by the training partner; one kit per climber group (19A)
P5	Aluminium welding not available near the partner	AC TIG is less common than stick welding in rural workshops	Frames made far away or poorly welded	3	3	Fabrication partner	Find a shop with AC TIG before the build; weld test pieces first
P6	Users take the open design as certified equipment	Published drawings and a working-looking kit	Use without training or proof testing	2	5	Project lead	"Not certified equipment" on every document, drawing and the spine marking; drill card with the gates

Preliminary hazard analysis

Table 2. Hazards.

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
H1	Collar slides down the trunk under load	Friction on wet bark below what the lock needs; slack chain; bark sloughing off	Victim and rescuer dropped or jerked; fall from height	5	2	Self-locking geometry (factor 1.54 on assumed friction); pads and sleeve; friction and proof tests before any use on a person	Medium until the friction is measured
H12	Collar slips on a leaning trunk	Frame on the adverse side of a trunk leaning 20 to 30 deg	Collar slides (as H1)	5	2	Lean limit of 20 deg (PR-003) and a drill rule that sets the frame on the favourable side (TKB-DEC-001 O8, decided 2026-10-25); friction test on a leaning section	Medium
H2	Collar loaded upward	Collar set below the victim's attachment, or the victim swung above it	Single top chain lifts out of grip; collar slips	5	2	Drill rule: collar at least 300 mm above the attachment (16A); training; misuse test at TRL 4	Medium
H3	Shock load when the victim's feet are freed	Slack left in the rope	2.6 to 3.0 kN at 30 mm of slack, 4.3 kN at 100 mm	5	2	Drill: take all slack through the descender first; frame covers a 30 mm drop (TKB-STM-001 S2); higher proof load (O7)	Medium
H4	Rescuer loses the brake strand	Fatigue, a blow, fainting	Uncontrolled descent	5	2	Auto-locking descender with anti-panic function (PR-011); brake carabiner keeps the hand force near 80 N	Low
H5	Victim slips out of the triangle or hangs head down	Wrong fitting; unconscious victim	Fall; airway risk	5	2	Pre-rigged triangle and chest sling on one carabiner (PR-012); drill with a dummy	Medium
H6	Frame member or weld breaks	Poor TIG weld; overload beyond about 5.0 kN (4.5 kN with an unequal chain split); a slack drop of 100 mm (4.3 kN); fatigue after many uses	Fall	5	2	Welding procedure and inspection (BLD-001); proof test of every frame at 3.5 kN and a sample frame tested to failure, the stronger frame the fallback below 5:1 (TKB-DEC-001 O7, decided 2026-10-25); 55 mm cleat web (D-S2)	Medium (strength below the low end of practice in the worst case)
H7	Chain or keeper		Collar released	5	2	Ball-lock pin through keeper, cheeks and web;	Low

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
	released at height	Pin not fully in; keeper not over both slots				fixed end tethered (PR-018); buddy check in the drill	
H8	Dropped kit or tool strikes helpers	Parts dropped while rigging; the bag hauled up	Head injury below	3	3	Tethered fixed end, lanyarded pin; helpers stand clear while the bag moves; helmets	Low
H9	Harm to the palm	Bare chain links on soft bark; over-tight wrap	Bark marks; permanent wounds (palms do not heal)	2	2	Pads at 1.9 MPa, sleeve 0.6 MPa, links mark about 0.3 mm; no spikes or screws (PR-008)	Low
H10	Heat at the hand	Long fast lowering	Burn to the brake hand	2	1	Brake strand warms well under 1 K; gloves; lower at about 0.3 m/s (TKB-STM-001 S29)	Low
H11	Victim's condition after hanging head down	Long suspension before help	Medical harm	5	3	Kit shortens the wait to minutes; emergency services always called; medical assessment after every rescue (soft, non-clinical wording)	Medium

Design FMEA (first pass, P3)

Table 3. Failure modes of option A. RPN = severity x occurrence x detection.

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F1	Rubber bark pads	Grip and spread the load on the bark	Slip on wet bark	Collar slides (H1)	Friction lower than 0.5 assumed; glazed or oily rubber	5	3	4	60	Friction test on wet trunk sections (BLD first check 1); pad compound to choose from test; replace glazed pads
F2	Chain sleeve	Grip at the back of the trunk, protect the bark	Slides or wears through	Lower grip; bark marked	Friction lower than 0.4; abrasion	4	2	3	24	Include in the friction test; replace when cut through (TKB-DWG-109)
F3	Cleat web (spine top)	Hold the chain links in the slots	Ligament yields; slot faces wear or dent	Chain link slips out of its slot	All of the chain's pull on one slot (11.1 kN at 2.5 kN); steel on aluminium bearing at 92 MPa	5	2	3	30	55 mm web (D-S2, factor 1.21 in the worst split); proof test on a 450 mm section with the chain pulled snug from one end; inspect slot faces (TKB-STM-001 S9, S10); O7
F4	Bar to spine welds	Carry the pad force into the spine	Crack in the weld or heat-affected zone	Bar bends or breaks	Poor AC TIG weld; porosity	5	2	3	30	Weld test pieces; visual and dye check of every frame; proof test
F5	Keeper and lock pin	Keep the links in the slots	Pin not fully home, or lost	Link lifts out under a shock	Bark debris in the pin hole; lanyard cut	5	2	2	20	Buddy check; pin click; lanyard to the keeper
F6	Tether link and tether hole (D-S1)	Keep the fixed end on the frame	Web strip bent; connecting link pin lost	Fixed end free when the keeper is off	Dropped frame; unpeened pin	2	2	2	8	Peen the pin; inspect the strip (TKB-DEC-001 item 14)
F7	Load eye and doubler rings	Carry the load carabiner	Tear-out or wear of the eye	Load released	Overload far beyond 2.5 kN (factor 16 on tear-out)	5	1	2	10	Round the bore edges; inspect for grooves
F8	Auto-locking descender (bought)	Control and stop the lowering	Fails to lock; wrong reeving	Uncontrolled descent	Wrong rope diameter; reeved backwards	5	2	2	20	Buy for 10.5 to 11.5 mm rope; maker's reeving check in the drill card
F9						3	2	2	12	

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
	Brake carabiner (bought)	Turn the brake strand, lower the hand force	Gate open, cross-loaded	Higher hand force; carabiner damage	Gate not screwed shut					Buddy check of every gate
F10	Rope (bought)	Carry and lower the person	Cut or abraded	Fall	Rope over a sharp edge; UV aged	5	2	2	20	Rope runs clear of the frame edges (model checks); retire on the maker's rules (TKB-STM-001 S26)
F11	Evacuation triangle and chest sling (bought)	Hold the victim head up	Wrongly fitted; victim slips out	Victim falls or hangs badly	Haste at height	5	2	3	30	Pre-rigged set (17B); dummy drill; training
F12	Frame (bare 6082-T6)	Whole-kit structure	Corrosion at steel contacts	Loss of section over years	Galvanic contact with steel chain, wet storage	3	2	3	18	Store dry; barrier paste at the rivet nuts; inspect after each season (TKB-STM-001 S25)

The highest RPN (F1, 60) is the friction of the pads on wet bark, the same unknown that leads the STEM check (S3) and the feasibility assessment; it is the first TRL 4 test.